

CLOUDCOSTIQ

Multi-Cloud Cost & Resource Optimization Analytics

Project Report

Project Domain	Cloud Infrastructure & DevOps Analytics
Problem Statement	Analyze AWS/Azure/GCP billing and usage data to identify underutilized resources and recommend cost-optimization actions.
Analysis Period	January–December 2025
Tools Used	Excel, MySQL, Python (Pandas), Power BI and GitHub
Dashboard	4-page interactive Power BI dashboard

1. Executive Summary

CloudCostIQ is an end-to-end data analytics project built to understand cloud spending and resource utilization across AWS, Azure and GCP. The main purpose is simple: find resources that are costing money without being used efficiently, quantify the possible savings, and present the findings in a way that a business or IT team can act on.

The project follows a practical Excel → MySQL → Python → Power BI workflow. The dataset contains monthly cloud billing and usage information for 2025, including cost, CPU and memory utilization, storage, network usage, running hours, resource status and potential savings. The final Power BI dashboard turns these details into four views: Cloud Cost Overview, Resource Utilization, Cost Optimization, and Resource Details & Recommendations.

In the final dashboard view, the project reports a current cloud cost of approximately ₹7.68 Cr, potential savings of ₹95.59 Lakh, an optimized cost of approximately ₹6.73 Cr, and a potential savings rate of 12.44%. The dashboard also identifies 120 resources in total and 43 underutilized resources for review.

2. Problem Statement

Cloud environments can grow quickly as teams create compute, storage, database and other services. When resource usage is not reviewed regularly, organizations may continue paying for capacity that is not being fully utilized. This project addresses that problem by combining billing data with utilization metrics and turning the analysis into clear optimization opportunities.

3. Project Objectives

- Measure total cloud spending and potential savings.
- Compare AWS, Azure and GCP spending patterns.
- Identify underutilized resources using utilization indicators.
- Understand how CPU, memory, storage and network usage vary across providers and environments.
- Find services and departments with higher cost or higher savings opportunities.
- Highlight resource-level optimization priorities for further review.
- Build an interactive Power BI dashboard for management-friendly analysis.

4. Dataset & Technology Stack

The supplied cloud billing dataset contains monthly resource-level records for AWS, Azure and GCP. The main analytical fields include Resource_ID, Cloud_Provider, Service_Name, Resource_Type, Region, Department, Environment, Instance_Type, Billing_Date, Monthly_Cost_INR, CPU_Utilization_Pct, Memory_Utilization_Pct, Storage_Usage_GB, Network_Usage_GB, Running_Hours, Requests_Count, Resource_Status, Utilization_Status, Potential_Saving_INR, Optimized_Cost_INR, Month, Year, Utilization_Score_Pct, Cost_per_Running_Hour_INR and Savings_Pct.

Tool	Role in the project	Output
Excel	Initial data preparation and review	Structured billing workbook and cleaned working data
MySQL	Database storage and SQL analysis	Aggregations, comparisons and optimization queries
Python / Pandas	EDA, validation and analytical calculations	Summary metrics, cleaning checks and supporting analysis
Power BI	Interactive visualization	Four-page management dashboard
GitHub	Project versioning / portfolio sharing	Project repository and documentation

5. End-to-End Methodology

Phase 1 – Excel / Data Preparation

The data was reviewed for missing values, inconsistent text, duplicate records, invalid utilization values, mixed formats and currency/percentage representations. Text standardization and numeric validation were treated as important because even small inconsistencies can change dashboard counts and groupings.

Phase 2 – MySQL

The prepared data was loaded into MySQL and analyzed using SQL. Queries were used to calculate total cost, provider and service comparisons, monthly trends, underutilized-resource counts, potential savings and optimization impact.

Phase 3 – Python

Python and Pandas were used to inspect the dataset, clean utilization-status text, validate summary metrics and support monthly and utilization analysis. The analysis also helped confirm the numbers used in the Power BI dashboard.

Phase 4 – Power BI

The final dashboard was designed as four pages. Slicers for date, cloud provider, region, environment and department allow the user to move from an overall management view to resource-level investigation.

6. SQL Analysis

MySQL was used to answer the main business questions from the billing table. The analysis included:

- Total cloud cost and potential savings.
- Cost comparison across AWS, Azure and GCP.
- Monthly cloud-cost trend.
- Underutilized-resource identification.
- Potential savings by cloud provider.
- Potential savings by service.
- Potential savings by department.
- Optimized cost and savings percentage.

A key calculation used in the project is: $\text{Optimized Cost} = \text{Monthly Cost} - \text{Potential Saving}$. This makes the savings story easy to explain and keeps the KPI cards consistent across SQL, Python and Power BI.

7. Python Analysis

Python was used to make the analysis reproducible and to validate the numbers before dashboard presentation. Pandas was the main library for loading, cleaning, filtering and grouping the data.

- Checked dataset shape, columns, data types, missing values and duplicates.
- Standardized Utilization_Status text using trimming and case normalization.
- Created ordered month analysis for January–December.
- Calculated current cost, potential savings, optimized cost and savings percentage.
- Separated underutilized records for deeper analysis.
- Reviewed provider, department and utilization patterns using grouped summaries and charts.

8. Power BI Dashboard

The Power BI report was designed to tell a simple story: how much the cloud costs, how efficiently resources are being used, where savings exist, and which resources need attention.

Page 1 – Cloud Cost Overview

This page provides the management-level view of cloud spend. KPI cards show total cloud cost, total resources, underutilized resources and potential savings. Supporting visuals compare provider cost, monthly cost, service cost and department cost.

Final dashboard KPIs: ₹7.68 Cr current cost; 120 resources; 43 underutilized resources; ₹95.59 Lakh potential savings.

Page 2 – Resource Utilization

This page focuses on operational efficiency. It compares CPU and memory utilization across providers and environments, shows cost versus CPU utilization, and breaks resources into utilization-status groups.

Final dashboard KPIs: average CPU 48.96%; average memory 49.26%; average storage usage about 1K GB; average network usage about 2K GB. The page identifies 43 underutilized resources.

Page 3 – Cost Optimization

This page moves from cost measurement to savings opportunities. It shows current cost, potential savings, optimized cost and savings percentage, followed by provider, service and department savings analysis.

Final dashboard KPIs: ₹7.68 Cr current cost; ₹95.59 Lakh potential savings; ₹6.73 Cr optimized cost; 12.44% potential savings.

Page 4 – Resource Details & Recommendations

This page is the action-oriented view. The detailed table allows individual resources to be reviewed using cost, CPU, memory, running hours, status and potential savings. Conditional formatting makes high-risk or high-opportunity rows easier to spot.

Optimization priority cards: 12 high-cost/low-CPU resources; 9 high-cost/low-memory resources; 7 with both low utilization indicators; 28 total priority resources.

9. Key Business Insights

- The dashboard indicates that 43 of the 120 resources are currently classified as underutilized, so a meaningful portion of the resource estate deserves review.
- Average CPU and memory utilization are both around 49%, suggesting that overall capacity usage is moderate rather than consistently high.
- Potential savings of ₹95.59 Lakh represent 12.44% of the current cloud cost in the final dashboard view. This is a potential opportunity, not a guaranteed realized saving.
- Provider, service and department views show that optimization should not be treated as a single cloud-wide action; the opportunity varies by workload and organizational area.
- The resource-level page helps move from a high-level KPI to a specific action by showing the resource, cost, utilization and potential saving together.

10. Next Best Actions

- Review and right-size underutilized resources, especially resources with high cost and low CPU or memory utilization.
- Prioritize high-cost, low-utilization resources where the potential saving is large enough to justify technical review.
- Monitor CPU, memory, storage, network usage and running hours regularly so that unused or oversized capacity is identified earlier.

These actions should be validated by the responsible cloud or engineering team before implementation because utilization alone does not capture application criticality, redundancy requirements, performance targets or future demand.

11. Business Recommendations

Recommendation	Practical approach
Right-size compute workloads	Review VM/compute capacity where CPU and memory remain consistently low. Reduce instance size only after checking application performance requirements.
Review idle resources	Investigate resources with very low running hours or usage. Stop, schedule or retire them where business ownership confirms they are no longer required.
Optimize storage	Review storage services for unused, oversized or infrequently accessed capacity and consider appropriate lifecycle or tiering policies.
Use provider-level monitoring	Compare AWS, Azure and GCP cost and utilization patterns regularly instead of reviewing providers independently.
Track savings after action	Create a monthly before-versus-after check for cost, utilization and realized savings so that optimization remains measurable.

12. Conclusion

CloudCostIQ demonstrates how a cloud billing dataset can be converted into a practical analytics solution using a familiar data-analyst workflow. Excel supports preparation, MySQL supports structured analysis, Python supports validation and exploration, and Power BI turns the results into an interactive story.

The final dashboard brings cost and utilization together rather than looking at either metric alone. The main outcome is not simply a cost number; it is a prioritized view of where cloud resources should be reviewed and where potential savings may exist. With regular monitoring and validation by cloud owners, the same framework can be extended to future months and used as an ongoing cloud-cost management process.

13. Skills Demonstrated

- Excel data cleaning and preparation
- MySQL database creation and SQL aggregation
- Data quality checking and standardization
- Python / Pandas exploratory data analysis
- Power BI data modeling, DAX measures and interactive dashboards
- KPI design, conditional formatting and dashboard storytelling
- Business insight generation and recommendation writing

14. Technical Appendix – Key Calculations

Metric	Definition
Total Cloud Cost	SUM(Monthly Cost INR)
Potential Savings	SUM(Potential Saving INR)
Optimized Cost	Total Cloud Cost – Potential Savings
Savings %	Potential Savings ÷ Total Cloud Cost
Total Resources	Distinct count of Resource_ID
Underutilized Resources	Distinct count of Resource_ID where Utilization_Status = Underutilized